Structured summary: Comparative Statistical Power of N-of-1 Trials versus Randomized Controlled Trials: A Simulation Study of Prazosin for PTSD Nightmares

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Contents

This document is a structured, section-by-section summary of the key points argued in the companion manuscript, "Comparative Statistical Power of N-of-1 Trials versus Randomized Controlled Trials: A Simulation Study of Prazosin for PTSD Nightmares." It is provided as a supplementary reading aid and does not stand on its own; all notation, derivations, simulation detail, and citations are in the main manuscript.

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1 Introduction

N-of-1 trials have emerged as a methodological complement to RCTs for settings where between-patient variability is clinically central.

- Traditional RCTs establish population-level effects but do not capture between-patient variability in treatment response.
- Precision medicine approaches are motivated by substantial between-patient variance in treatment response.
- The statistical properties of N-of-1 designs relative to RCTs across realistic effect-size and carryover regimes remain incompletely characterized.

Aggregated N-of-1 trials reduce between-patient confounding while supporting both individual and population-level treatment-effect inference.

- Each patient serves as their own control in repeated crossover periods, eliminating between-patient variability from the estimator.

- Aggregation across patients enables population-level inference while simultaneously informing individual treatment decisions.
- The design provides individual-level treatment-effect estimates unavailable from parallel-group comparisons.

1.1 Methodological Considerations in N-of-1 Trial Design

The statistical analysis of N-of-1 trials requires explicit attention to carryover, period effects, and serial correlation within patients.

- Carryover effects from prior treatment periods are particularly critical in crossover designs and can affect validity and power.
- Temporal trends and within-patient autocorrelation present analytical challenges absent from parallel-group designs.
- Appropriate handling of these nuisance features is a prerequisite for unbiased estimation.

Population-level inference from N-of-1 series requires meta-analytic methods that account for both within- and between-patient variability.

- Random-effects meta-analysis methods such as DerSimonian-Laird appropriately combine individual treatment-effect estimates.
- Both within-patient and between-patient variance components must be incorporated to avoid bias.
- The aggregation strategy chosen influences the efficiency and interpretability of the combined estimate.

1.2 Sample Size and Power Considerations

The statistical efficiency of N-of-1 trials relative to RCTs is context-dependent and has not been comprehensively evaluated across effect sizes and carryover scenarios.

- Sample size in N-of-1 designs involves the joint optimization of patient count, crossover periods, and within-to-between variability ratio.
- Prior simulation studies suggest N-of-1 designs achieve adequate power at smaller total sample sizes, particularly for moderate effects with substantial individual variability.
- The relative efficiency advantage and its dependence on carryover regime remain open empirical questions.

1.2.1 Analytic Frameworks for Power in N-of-1 Designs

N-of-1 statistical power depends on a fundamentally different variance decomposition than parallel-group RCTs, as between-patient variance is eliminated from the treatment-effect estimator.

- The relevant uncertainty sources are the within-patient between-period variance and residual measurement error.

- This variance reduction relative to RCTs underpins the N-of-1 efficiency advantage.
- The decomposition implies that power depends on within-patient noise and the number of crossover cycles, not between-patient heterogeneity.

1.2.1.1 Variance decomposition and the paired-cycles model **Senn formalized sample-size planning for N-of-1 series under a Normal-Normal mixture model where cycle-level differences follow a random-effects structure.**

- The model decomposes within-cycle treatment differences into a patient-level random effect and within-patient residual noise.
- τ^2 captures between-patient variance in individual treatment effects and $2\sigma^2$ the within-patient within-cycle variance.
- The paired-cycles structure implies that power depends on the ratio of within-patient noise to patient heterogeneity.

The factor of 2 in the variance arises because each within-cycle difference is computed from two independent within-period observations.

- Δ is the population mean treatment effect and τ^2 is the between-patient variance of individual treatment effects.
- $2\sigma^2$ is the within-patient within-cycle variance of paired differences.
- This parameterization identifies the two variance components that determine sample size requirements.

The grand mean estimator variance $\mathbf{Var}(\hat{\Delta}) = (\tau^2 + 2\sigma^2/k)/n$ reveals the trade-off between number of patients and cycles per patient.

- Increasing k reduces only the within-patient $2\sigma^2/k$ component, while increasing n reduces the entire expression.
- When between-patient heterogeneity τ^2 dominates, additional patients yield greater precision than additional cycles.
- This trade-off is central to sample-size planning for N-of-1 series and has no simple universal optimum.

1.2.1.2 Closed-form power for fixed- and random-effects analyses **Senn distinguished random-effects and fixed-effects inferential goals with distinct power formulas and degrees-of-freedom structures.**

- The random-effects analysis targets the population treatment effect Δ and yields a one-sample t-test on the n patient-level means.
- The noncentrality parameter for the random-effects case is $\lambda_{\mathrm{RE}} = \Delta/\sqrt{(\tau^2 + 2\sigma^2/k)/n}$ with $n-1$ degrees of freedom.
- The random-effects analysis estimates the patient-level variance directly from the n observed summary measures.

The fixed-effects analysis excludes the treatment-by-patient interaction from the

variance estimator and uses $n(k-1)$ degrees of freedom from pooled within-patient sums of squares.

- The relevant variance of $\hat{\Delta}$ is $2\sigma^2/(nk)$, estimated with $n(k-1)$ degrees of freedom.
- The noncentrality parameter is $\lambda_{\mathrm{FE}} = \Delta/\sqrt{2\sigma^2/(nk)}$.
- Standard sample size software assuming $n-1$ degrees of freedom slightly underestimates power in the fixed-effects case when $k > 2$.

Senn and Julious provided a complementary tutorial on computing pooled within-patient standard deviations and constructing shrinkage estimators for individual patient effects.

- The pooled within-patient SD is estimated as $\hat{\sigma}_w = \sqrt{\sum_i \mathrm{SS}_i / \sum_i (m_i - 1)}$.
- Shrinkage estimators, that is best linear unbiased predictors (BLUPs), for individual treatment effects can be derived from the random-effects model.
- These practical tools bridge the theoretical variance decomposition and applied N-of-1 analysis.

1.2.1.3 The n-versus-k trade-off in series of N-of-1 trials **Zucker et al. showed that the marginal benefit of additional crossover cycles diminishes rapidly once within-patient noise falls below between-patient heterogeneity.**

- When $\tau^2 \gg \sigma_w^2$, additional patients yield greater precision than additional cycles.
- Conversely, when within-patient noise is large relative to heterogeneity, repeated cycles on the same patient remain valuable.
- This result motivates joint optimization of n and k in N-of-1 series planning.

Arends et al. extended Zucker et al.'s results to derive explicit sample-size formulas for N-of-1 series analyzed via random-effects meta-analysis.

- Explicit tables jointly determine n and k as a function of true effect Δ, within-patient variance σ_w^2, and between-patient heterogeneity τ^2.
- These formulas provide a closed-form foundation for N-of-1 trial planning in the absence of carryover or serial correlation.
- The present simulation study tests the closed-form predictions under realistic violations of the simplifying assumptions.

1.2.1.4 Complications motivating simulation **Closed-form power solutions assume negligible carryover, an assumption that may not hold when pharmacokinetic washout between periods is incomplete.**

- Partial carryover attenuates the effective treatment contrast and shifts variance components in ways not captured by balanced-design formulas.
- Prazosin's neuroadaptive effects on sleep architecture may produce precisely such partial washout.

- These violations motivate Monte Carlo simulation as a complement to closed-form sample-size planning.

Ignoring within-patient serial correlation can inflate Type I error or distort power estimates in N-of-1 trials.

- Wang and Schork demonstrated via AR(1) likelihood-ratio methods that serial correlation can substantially reduce power.
- More frequent period alternation partially mitigates this power loss at the cost of practical feasibility.
- Designs with few, long treatment periods are most vulnerable to power loss from unmodeled serial correlation.

Several additional violations of the balanced paired-cycles assumptions further motivate Monte Carlo simulation for realistic N-of-1 power evaluation.

- Practical N-of-1 trials may feature unbalanced designs, missing periods, or adaptive stopping rules.
- Count data outcomes violate the normality assumption underlying closed-form formulas.
- Joint optimization of n and k for designs targeting both individual and population-level inference has no simple closed form.

These accumulated complications collectively motivate the Monte Carlo simulation approach, which permits power evaluation under realistic combinations of carryover, variance structure, and design imbalance.

- Simulation accommodates carryover, serial correlation, and design imbalance simultaneously.
- ADEMP-compliant simulation reporting with Monte Carlo standard errors provides quantified uncertainty for all performance measures.
- The present study fills the gap left by closed-form analyses that assume idealized conditions.

1.3 Single-Case Experimental Design Framework

N-of-1 trials represent a specific application of single-case experimental designs whose methodological rigor has evolved to support valid causal inferences from individual-level data.

- SCED methodology has been extensively developed in behavioral and educational research contexts.
- Analytical approaches for SCEDs now include tools for valid causal inference from individual trajectories.
- Methodological standardization from the SCED literature provides a foundation for rigorous N-of-1 trial analysis.

Recent SCED advances emphasizing proper randomization, adequate baseline

measurement, and appropriate statistical analysis have strengthened the evidence base for N-of-1 trials as a complement to group designs.

- Randomization, adequate baseline, and appropriate analysis are now recognized as essential SCED quality criteria.
- These advances position N-of-1 trials as methodologically credible complements to parallel-group designs.
- Standardised SCED reporting and analysis guidelines reduce implementation barriers.

1.4 Clinical Application: Prazosin for PTSD

Prazosin for PTSD nightmares provides an instructive clinical context for N-of-1 methodology given substantial individual response variation and the chronic fluctuating nature of PTSD symptoms.

- Prazosin has demonstrated efficacy in multiple RCTs for reducing trauma-related nightmares via α_1-adrenergic blockade.
- Individual response patterns vary considerably, making individual-level inference an important clinical goal.
- The PTSD/nightmare context provides empirically grounded simulation parameter calibration.

Prazosin's pharmacokinetics, including its relatively short half-life, make it suitable for crossover designs while necessitating careful consideration of carryover effects.

- The short half-life facilitates crossover trial design by limiting washout duration.
- Previous studies document both efficacy and substantial individual variability of prazosin response.
- This empirical foundation justifies the simulation parameters employed in the present study.

1.5 Regulatory and Implementation Considerations

Increasing regulatory recognition of N-of-1 trials has highlighted the need for robust methodological guidelines and statistical frameworks.

- Regulatory agencies and research institutions have acknowledged N-of-1 trials as a valid evidence-generation approach.
- Professional organizations have developed comprehensive recommendations for N-of-1 trial design and implementation.
- Specialised software tools and reporting standards have emerged to support N-of-1 trial analysis.

1.6 Prior Simulation Comparisons of N-of-1 and Parallel Designs

The comparative efficiency of N-of-1 and parallel-group designs has been examined in several simulation studies that form the immediate methodological precedent for the present work.

- A companion literature review provides a structured summary of the key findings.
- Direct head-to-head simulation comparisons, analytical complications, closed-form frameworks, and recent design refinements are each addressed.
- Key findings from prior studies are summarized in the following subsections.

1.6.1 Direct head-to-head simulation comparisons

Blackston et al. reported that aggregated N-of-1 designs achieve higher power than both parallel RCTs and two-period crossovers at matched sample sizes, and that unmodeled carryover inflates N-of-1 Type I error.

- Using 5,000 simulated trials with three cycles, N-of-1 designs exceeded 80% power at $n = 30$ while RCTs required $n = 150$.
- Type I error under N-of-1 inflated to 15–25% when moderate-to-strong carryover was present but not modeled.
- Patient-level random effects are recoverable in N-of-1 but not in parallel RCTs, which provide only one observation per participant.

Carrozzo et al. found that a meta-analytic N-of-1 procedure reached 80% power with fewer participants than two-arm parallel or two-period crossover designs across most heterogeneity and carryover settings.

- The simulation varied between-subject heterogeneity and carryover magnitude for a cardiovascular digital-health intervention.
- Both a fixed-n meta-analysis and a sequential-aggregation procedure were evaluated.
- The sequential variant crossed the 80% power threshold earlier than the fixed-n procedure.

Hendrickson et al.'s simulation of aggregated N-of-1 designs for predictive-biomarker validation is the methodological ancestor of the biomarker-interaction framework used in the present sensitivity analyses.

- Four designs were compared at 1,000 replicates per scenario: open label, blinded discontinuation, traditional crossover, and a hybrid.
- The hybrid design provided power close to the traditional crossover while preserving an initial open-label titration phase.
- The hybrid showed smaller power erosion under carryover than the other designs.

1.6.2 Analytical complications and misspecification

Wang and Schork extended power analysis to cases with AR(1) serial correlation, heteroscedastic responses, and washout periods, demonstrating that ignoring serial correlation substantially reduces power.

- More frequent period alternation partially mitigates serial-correlation power loss at the cost of practical feasibility.
- Washout periods reduce effective sample size in proportion to the carryover half-life relative to the period length.
- These findings motivate the `corCAR1` residual correlation structure used in the present N-of-1 analysis model.

Gaertner et al. simulated aggregated N-of-1 trials under directed acyclic graphs encoding carryover structures, finding that a carryover-adjusted parametric model is unbiased while other methods show bias under carryover.

- Linear mixed models, Bayesian networks, G-estimation, and a carryover-adjusted model were compared.
- All methods perform well without carryover; under carryover, only the carryover-adjusted model is unbiased.
- This work quantifies the cost of mismatched carryover specification within the N-of-1 analytic family.

1.6.3 Closed-form and semi-analytical frameworks

Senn identified five sample-size planning criteria with closed-form solutions for N-of-1 series and characterized the n-versus-k trade-off through the variance decomposition $\mathbf{Var}(\hat{\Delta}) = (\tau^2 + 2\sigma^2/k)/n$.

- The decomposition makes explicit that increasing k reduces only the within-patient $2\sigma^2/k$ term, while increasing n reduces the full expression.
- Zucker, Ruthazer, and Schmid compared summary and individual-participant-data meta-analyses on a published N-of-1 series and characterized the variance structure determining relative efficiency.
- These closed-form results provide benchmarks against which the present simulation findings can be assessed.

1.6.4 Recent design refinements

Recent work extends N-of-1 design analysis in three directions: sequential monitoring, per-trial monitoring with random-effects combination, and generalized pairwise comparisons.

- Jiang et al. proposed sequential monitoring rules for single-person and multi-person N-of-1 trials with sample-size calculations in an Alzheimer's disease application.

- Selukar et al. found that Type I error can be substantially inflated under a small number of planned blocks but reaches nominal rates with alpha-spending corrections.
- Giai et al. reported that individual-level Net Benefit estimation requires very long observation series to achieve adequate power.

1.6.5 Summary

The N-of-1 efficiency advantage is largest when between-patient heterogeneity is non-negligible, carryover is absent or correctly modeled, and within-patient serial correlation is accounted for.

- The advantage erodes when carryover exceeds the period length, when strong AR(1) correlation is ignored, or when the number of cycles per patient is small.
- Type I error is more sensitive to analytic misspecification under N-of-1 than under parallel designs.
- Biomarker-treatment interaction analyses have qualitatively different power profiles than average-effect analyses.

1.7 Objectives

The primary objective is to compare statistical power of aggregated N-of-1 hybrid trials versus parallel-group RCTs, with secondary objectives examining carryover effects, period-exclusion robustness, and conditions favoring N-of-1.

- The prazosin/PTSD application provides a clinically calibrated reference for simulation parameter specification.
- Secondary objectives include evaluating carryover half-life impact and the robustness of the period-exclusion strategy.
- The conditions under which the N-of-1 approach is preferred over parallel-group designs are explicitly characterized.

2 Methods

We follow the ADEMP framework of Morris, White, and Crowther, organizing the simulation study around Aims, Data-generating mechanisms, Estimands, Methods, and Performance measures.

- ADEMP provides a structured reporting standard for Monte Carlo simulation studies.
- Pre-registration of the parameter grid fixes all primary and sensitivity cells before production runs.
- Each component of ADEMP is described in turn in the following subsections.

2.1 Estimands and pre-registration

The primary estimand is the average treatment-effect coefficient β_D from the N-of-1 linear mixed-effects model, with statistical power $\pi = \Pr(p < 0.05)$ as the

primary inferential output.

- Secondary estimands include the bias of $\hat{\beta}_D$ relative to the calibrated true value and the empirical Monte Carlo standard error.
- All performance measures are reported with binomial-proportion Monte Carlo standard errors for proportions and standard MCSE for continuous quantities.
- The full ADEMP plan is pre-registered at `analysis/scripts/treatment-main-effect/00-ademp-pre-r`

2.2 Simulation design and data-generating mechanism

Both designs operate on the same patient pool within each replicate, following the Morris et al. paired-comparison discipline to reduce Monte Carlo variance of power differences.

- Each replicate draws $N = 35$ patient-level baseline nightmare frequencies from a Normal distribution calibrated to yield $I^2 \approx 50\%$.
- The shared pool drives the N-of-1 and RCT arms independently within each replicate.
- The within-replicate covariance between power estimates is exploited when reporting paired power differences.

The N-of-1 arm simulates 8-period balanced crossover trials with continuous-time AR(1) within-patient residuals and exponential carryover decay.

- Within-patient residuals follow a zero-mean multivariate normal with $\mathrm{Cov}(\varepsilon_j, \varepsilon_k) = \sigma_w^2 \rho^{|t_j - t_k|}$.
- Carryover from a drug period decays as $\delta_c = \beta_D \exp\{-\ln(2)L/t_{1/2}\}$, with $t_{1/2} = 3$ days at the primary cell.
- The true effect grid is $\beta_D \in \{0, -0.5, -1.0, -2.0\}$ nightmares per week.

The parallel-group RCT arm allocates the same 35 patients 1:1 to treatment or placebo, with the endpoint drawn from a linear model using the shared-pool baseline.

- The endpoint model is $Y_i = \mu_0 + u_i + \beta_D Z_i + \varepsilon_i$, using the patient's baseline draw from the shared pool.
- This structure ensures both arms are exposed to the same between-patient heterogeneity.
- The ANCOVA analysis uses the shared-pool baseline as covariate.

Carryover sensitivity sweep S1 varies the pharmacokinetic half-life across five values at fixed true effect $\beta_D = -2.0$.

- Half-life values $t_{1/2} \in \{0.5, 1.0, 3.0, 5.0, 7.0\}$ days span a clinically plausible range.
- The sweep tests the robustness of the period-exclusion strategy to half-life misspecification.
- Fixed effect size isolates the carryover half-life as the sole varying parameter.

2.3 Statistical analysis models

The N-of-1 arm is analyzed with a linear mixed-effects model with patient random intercept and `corCAR1` residual correlation, tested via Satterthwaite-corrected Wald t-statistic.

- The formula `Sx ~ Dbc + t_wk` with `corCAR1(form = ~t_wk|ptID)` fits continuous-time AR(1) residual correlation.
- The Satterthwaite denominator degrees of freedom correct the small-sample anti-conservatism observed in preliminary runs at $N = 35$.
- The null hypothesis $H_0 : \beta_D = 0$ is tested at the two-sided $\alpha = 0.05$ level.

The parallel-group RCT arm is analyzed by ANCOVA with the shared-pool baseline as covariate, using exact t_{N-3} degrees of freedom.

- The analysis model is `lm(endpoint ~ trt + baseline)`.
- Treatment coefficient and 95% CI are extracted from the usual t-distribution with $N - 3$ degrees of freedom.
- The ANCOVA covariate is the patient's baseline nightmare frequency from the shared simulation pool.

2.4 Simulation size and random-number discipline

We pre-specified $n_{\mathbf{sim}} = 2{,}000$ replicates per cell, applied uniformly to all primary and sensitivity cells, yielding MCSE below 1.5 percentage points for both power and Type I error.

- At $n_{\text{sim}} = 2{,}000$, the MCSE on a power estimate near 0.75 is approximately 1.0 percentage points.
- The MCSE on a Type I estimate near 0.05 is approximately 0.5 percentage points.
- Both are below the 1.5-percentage-point target adopted in the pre-registration.

Random-number discipline follows Morris et al. using L'Ecuyer-CMRG with a single master seed, with per-replicate seed states captured for exact reproduction.

- `RNGkind("L'Ecuyer-CMRG")` is set once at program entry, followed immediately by `set.seed(2024L)`.
- No `set.seed()` calls appear inside simulation functions.
- Per-replicate `.Random.seed` states are captured before each replicate to permit exact reproduction of anomalous results.

2.5 Software and reproducibility

All analyses were conducted in R 4.4+ with `nlme` for mixed-effects fitting and `parallel::mclapply` for multi-core execution, with full package dependencies captured in `renv.lock`.

- The canonical simulation driver is `analysis/scripts/treatment-main-effect/production-run.R`.

- Output is stored as `analysis/data/derived_data/04-mainfx-production.rds`.
- Full reproducibility requires the project's zzcollab Docker container with the recorded image hash.

3 script sourced above.

At the primary scenario, empirical power was substantially higher for the aggregated N-of-1 hybrid than for the parallel-group RCT, and the paired within-replicate power difference confirms this ranking is not attributable to simulation noise.

- N-of-1 power was 100.0
- The Morris paired design yields a within-replicate power difference of 0.325 with paired MCSE 0.0105.
- The paired MCSE is smaller than the square-root-sum of the two marginal MCSEs, reflecting variance reduction from the shared patient pool.

4 Morris-compliant summary table: all performance measures with MCSEs

The aggregated N-of-1 design achieved substantially higher power than the parallel-group RCT at the baseline simulation parameters.

- N-of-1 power was 100% (MCSE 0%) versus 67.6% for the RCT.
- The 95% Monte Carlo CI for the N-of-1 power is 100–100%.
- The power difference of 32.5 percentage points exceeds five combined MCSEs.

Standard error calibration and CI coverage are satisfactory for both designs: model SEs are well-calibrated and coverage tracks the nominal 95% level.

- The N-of-1 relative SE error is -0.021, confirming analytical SEs are well-calibrated.
- Coverage of nominal-95% CIs is 94.0
- Any N-of-1 coverage shortfall is attributable to systematic carryover bias in the simplified data-generating process (DGP).

4.1 Power curves

The N-of-1 hybrid achieves substantially higher power than the parallel-group RCT at every non-null effect size, and the power advantage grows as the true effect moves away from the null.

- At $\beta_D = -0.5$ nightmares per week, the RCT reaches only 10.4
- The N-of-1 empirical SE is calibrated to approximately 0.28 nightmares per week at the null cell, rising to approximately 0.34 at non-null cells where contaminated periods are excluded.

- Absolute power magnitudes should be interpreted in light of the simplified linear DGP, pending the full the mean-moderation architecture Gompertz production run.

4.2 Bias and coverage

Bias is negligible at all effect sizes for both designs, and the period-exclusion strategy successfully removes carryover contamination from the N-of-1 estimator.

- The N-of-1 bias is +0.011 at −0.5, +0.011 at −1.0, and +0.011 at −2.0.
- Carryover-contaminated periods contribute a residual treatment effect of approximately $0.199|\beta_D|$ at $t_{1/2} = 3$ days; excluding them removes this source of bias.
- Model SE calibration is good (relative error within ±2%) and coverage tracks the nominal level without further adjustment.

4.3 Power Across Effect Sizes and Carryover Scenarios

Statistical power varies substantially across the true-effect grid, with the N-of-1 hybrid retaining a power advantage at every non-null effect size.

- The aggregated N-of-1 hybrid retains a power advantage over the parallel-group RCT at every non-null cell.
- The magnitude of the advantage grows as the true effect moves away from the null.
- Both the effect-size sweep and the carryover half-life sweep are displayed in Figure **??**.

4.3.1 Effect Size Relationships

Across the effect-size grid, the N-of-1 hybrid outperformed the parallel-group RCT at the primary carryover half-life of 3 days, with the largest advantage at the moderate-effect cell.

- At $\beta_D = -1.0$ nightmares per week, the hybrid reached 83.0% power versus 23.0% for the RCT, a 60.0-percentage-point gap.
- The advantage was present at every non-null effect-size cell within the simulated range.
- The relative N-of-1-to-RCT power ratio was approximately 3.6× at the moderate-effect −1.0 cell.

4.3.2 Carryover half-life sensitivity (S1)

N-of-1 power, bias, and empirical SE are essentially constant across all five carryover half-life cells in the S1 sweep, confirming that period exclusion provides robustness to half-life misspecification.

- Once contaminated periods are excluded before analysis, the true half-life no longer enters the estimating equations.
- RCT power is also constant across S1 cells, as expected, since carryover does not affect a parallel-group design.

- A future sweep S11 will compare period exclusion against explicit carryover modeling when the true $t_{1/2}$ is available to the analyst.

The power curves did not cross within the simulated effect-size and carryover range, and period exclusion holds N-of-1 power essentially constant across the carryover half-life sweep.

- The N-of-1 advantage was positive at every cell of the combined effect-size and carryover grid.
- Period exclusion removes carryover-contaminated periods before fitting, so N-of-1 power does not decline with longer half-lives.
- Whether a crossover would emerge under explicit carryover modeling that retains contaminated periods is a question for sweep S11.

5 Display results

Both designs maintained appropriate Type I error control under the null hypothesis, with both 95% Monte Carlo confidence intervals including the nominal 5% level.

- The RCT design showed Type I error of 6.2% (95% CI 5.1–7.2%).
- The N-of-1 design showed Type I error of 5.1% (95% CI 4.1–6.1%).
- The observed power differences reflect genuine effect detection rather than inflated false-positive rates.

5.1 Between-patient variance and treatment heterogeneity

The random-intercept variance estimated by the `lme` model quantifies residual between-patient heterogeneity after adjusting for treatment, with a true between-patient SD of $\sigma_\tau = 1.5$ nightmares per week calibrated to yield an intraclass correlation coefficient (ICC) near 0.69.

- The DGP does not include a treatment-by-patient interaction; between-patient heterogeneity is purely in baseline nightmare level.
- Per-replicate estimates of $\hat{\sigma}_u^2$ are available in the `.rds` output for secondary heterogeneity analyses.
- The ICC of 0.69 represents the fraction of total variance attributable to patients rather than within-patient noise.

6 Create Morris et al. (2019) compliant summary table

The comprehensive simulation performance summary confirms essentially unbiased effect estimation for both designs, with the N-of-1 design demonstrating superior precision and higher power while maintaining appropriate Type I error control.

- Absolute bias was below 0.02 nightmares per week for both designs at all effect sizes.
- The N-of-1 design achieved lower empirical SE reflecting the within-patient comparison efficiency.
- Type I error was maintained at or near nominal levels, validating the power comparison.

6.1 Effect Size Distribution Comparison

The distribution of effect-size estimates illustrates the superior precision of the N-of-1 design, with both designs providing essentially unbiased estimates but N-of-1 showing a tighter distribution around the true value.

- The tighter N-of-1 distribution reflects efficiency gains from within-patient comparisons.
- Both designs are centered near the true effect, confirming negligible bias.
- The width difference between the two distributions is the primary driver of the observed power gap.

6.2 Optimal Design Conditions

The N-of-1 advantage was largest under four conditions: large true effect, carryover half-lives short relative to the period length, substantial between-patient baseline variability, and moderate within-patient measurement noise.

- At large true effects ($\beta_D = -2.0$), the hybrid saturates power while the RCT remains underpowered.
- Short carryover half-lives minimize the number of excluded periods, preserving effective sample size.
- Substantial between-patient variability ($\sigma_\tau = 1.5$ nightmares per week) is eliminated from the N-of-1 estimator, conferring a large efficiency advantage.

At the primary scenario, the aggregated N-of-1 hybrid achieved a rejection rate of 1.000 compared with 0.675 (MCSE 0.010) for the parallel-group RCT.

- True effect -2.0 nightmares per week, half-life 3 days, $N = 35$ per design, and $n_{\text{sim}} = 2{,}000$.
- The N-of-1 saturated power while the RCT remained substantially underpowered.
- This gap confirms the N-of-1 advantage in the clinically relevant large-effect regime.

No configuration in the simulated grid favored the parallel-group RCT; a crossover did not materialise because period exclusion holds N-of-1 power constant across the carryover sweep.

- One might expect the RCT to gain ground when carryover half-lives approach or exceed the period length with small effect sizes.
- The period-exclusion analysis holds N-of-1 power essentially constant across the carryover sweep, preventing a crossover.

- Whether a crossover emerges under explicit carryover modeling is addressed by the pre-registered sweep S11.

6.3 Carryover Modeling Strategy Comparison

The present production run analyzes the N-of-1 arm under period exclusion only; a head-to-head comparison with explicit carryover modeling is pre-registered as sweep S11 and deferred.

- Period exclusion removes carryover-contaminated placebo periods before the `lme` model is fitted, preventing bias at the cost of a reduced effective sample size.
- Explicit carryover modeling retains every period and includes a continuous carryover regressor, preserving sample size while separating treatment and carryover effects.
- The two strategies are not in binary opposition; the analysis model already carries a continuous drug-exposure term.

6.4 Carryover Decay Model Sensitivity Analysis

The carryover decay is modeled as exponential, and the comparison with linear and Weibull alternatives in Figure ?? motivates the planned decay-model sensitivity sweep.

- The three curves diverge appreciably within the seven-day period, suggesting that decay kinetics are a parameter worth interrogating directly.
- A vertical dotted line at seven days marks the end of the off-drug period.
- The analytic curves should be read as motivation for sweep S3 rather than as a completed multi-model power comparison.

The production run simulates power under the exponential decay model only; linear and Weibull decay-model comparisons are pre-registered as sweep S3 and deferred.

- At the primary cell, the exponential decay model yields N-of-1 power of 83.0
- Sweep S3, comparing power under exponential, linear, and Weibull decay, remains an open question.
- The robustness of power estimates to decay-model choice cannot be assessed from the present results alone.

6.5 Sample N-of-1 Trial Illustration

A representative individual N-of-1 trial illustrates the crossover pattern and carryover identification, with carryover-affected periods excluded from the primary analysis.

- The patient completed 8 periods in a randomized treatment sequence.
- Placebo periods immediately following a drug period were identified as carryover-affected and excluded.

- The figure demonstrates the practical mechanics of the period-exclusion strategy.

7 Display sample trial analysis results

The individual treatment-effect estimate from the representative trial, comparing 4 prazosin periods with 3 pure placebo periods, is close to the true effect size.

- The individual estimate of -2.34 nightmares per week is consistent with the true effect.
- This single-patient illustration demonstrates the period-exclusion procedure in practice.
- Per-period outcomes for prazosin, pure placebo, and carryover-affected placebo are shown in Table **??**.

8 Discussion

Aggregated N-of-1 trials can provide substantially higher statistical power than traditional parallel-group RCTs in clinical scenarios characterized by substantial individual variability and modest carryover.

- The simulations bear on clinical trial design, particularly in precision-medicine applications where between-patient variability is substantial.
- The findings extend prior theoretical and simulation work by providing empirical power estimates across a clinically calibrated parameter range.
- The efficiency advantage is quantified across a grid of effect sizes and carryover half-lives.

8.1 Statistical Efficiency of N-of-1 Designs

The observed power advantage of N-of-1 trials reflects the principle that within-subject comparisons are more efficient than between-subject comparisons when individual baseline differences are large relative to measurement error.

- Within-patient designs eliminate between-patient variance from the treatment-effect estimator.
- The findings extend Blackston et al. (2019) and Chen et al. (2020) by providing power estimates over a clinically calibrated parameter range.
- The power advantage was present at every non-null effect size examined.

The power advantage is approximately 21 percentage points at $\beta_D = -0.5$, 60 at -1.0, and 32 at -2.0 where the hybrid saturates, with practical implications for sample-size planning in recruitment-constrained settings.

- Efficiency gains of this magnitude translate into meaningful reductions in required sample size at fixed power.
- The advantage is particularly valuable in rare-disease contexts or where recruitment is otherwise difficult.

- The relative efficiency improvement is largest in the small-to-moderate effect regime where parallel-group RCTs are essentially underpowered.

8.2 Carryover Effects and Design Optimization

The results clarify that modest carryover (half-lives short relative to the period length) can be managed through appropriate analytical methods without substantially compromising N-of-1 power.

- Intuition might suggest that any carryover would compromise an N-of-1 design, but the findings show otherwise.
- Period exclusion provides power robustness across the simulated half-life range.
- This robustness is conditional on correctly identifying and excluding contaminated periods.

8.2.1 Comparison of Carryover Handling Strategies

The two carryover-handling strategies, period exclusion and explicit carryover modeling, are not in binary opposition; both are defensible and differ chiefly in how much off-drug information the analyst chooses to retain.

- Period exclusion prevents bias by removing contaminated observations, at the cost of reduced effective sample size.
- Explicit carryover modeling retains contaminated periods and includes a carryover term to separate treatment and residual carryover effects.
- A head-to-head comparison of the two strategies under matched seeds is pre-registered as sweep S11.

The present results characterize the period-exclusion strategy only; a quantitative comparison with explicit carryover modeling awaits the pre-registered sweep S11.

- Bias was negligible and coverage tracked the nominal level at every cell under period exclusion.
- Recent methodological guidance suggests explicit carryover modeling can preserve power while delivering unbiased estimates.
- Sweep S11 will provide the quantitative comparison on the present design.

8.3 Clinical Context and Generalizability

Prazosin for PTSD nightmares is a well-suited test case for N-of-1 methodology due to substantial individual response variation, short pharmacokinetic half-life, and a continuous outcome suitable for repeated assessment.

- Substantial individual variation in treatment response motivates individual-level inference.
- The relatively short half-life of prazosin facilitates crossover designs.

- Nightmare frequency is a continuous outcome well-suited for repeated within-patient assessment.

The generalizability of the findings to other clinical contexts depends on whether similar conditions obtain: high between-patient variability, moderate-to-large treatment effects, and manageable carryover.

- N-of-1 designs are most likely to demonstrate power advantages in these conditions.
- Therapeutic areas with smaller individual variability or larger carryover may show attenuated N-of-1 advantages.
- The present simulation provides a framework for evaluating applicability to specific clinical settings.

8.4 Methodological Advances and Implementation

Recent advances in SCED methodology, standardized reporting guidelines, and specialized software tools have reduced implementation barriers and improved methodological rigor for N-of-1 trials.

- CONSORT extension guidelines for N-of-1 trials provide a standardized reporting framework.
- Specialised software tools support N-of-1 meta-analysis.
- These developments strengthen the evidence base for N-of-1 trials as a methodological complement to parallel-group designs.

The integration of Bayesian approaches and adaptive design elements represents a promising direction for enhancing N-of-1 trial efficiency and clinical utility.

- Bayesian approaches are particularly well-suited for adaptive N-of-1 designs that adjust crossover periods based on accumulating data.
- The simulation framework developed here could readily be extended to evaluate Bayesian adaptive variants.
- Both topics merit further simulation study on designs of the kind examined here.

8.5 Implications for Precision Medicine

The efficiency advantage of N-of-1 trials aligns with the goals of precision medicine, providing both population-level evidence and individual-level treatment-effect estimates.

- N-of-1 trials can simultaneously support regulatory approval and individual clinical decision-making.
- The design directly supports the precision-medicine goal of optimizing treatment selection based on individual patient characteristics.
- The dual population-level and individual-level inference capacity distinguishes N-of-1 from parallel-group designs.

The present DGP places between-patient heterogeneity entirely in baseline level without a treatment-by-patient interaction, so the simulation does not itself estimate heterogeneity of treatment effect.

- Individual variation in treatment response is nonetheless plausible in this clinical setting.
- A treatment-by-patient interaction term would be a natural extension warranting further study.
- Individual patients may experience appreciably different magnitudes of benefit from prazosin.

8.6 Limitations and Considerations

Several methodological limitations should be borne in mind when interpreting the findings.

- The simulation assumed perfect compliance and no missing data.
- A single continuous outcome measure was examined.
- The carryover decay model was exponential only in the present production run.

8.6.1 Simulation Design Limitations

The simulation assumed perfect compliance and no missing data, which may not reflect real-world implementation challenges in N-of-1 trials.

- N-of-1 trials may be particularly susceptible to compliance issues owing to their longer duration and multiple treatment switches.
- Missing data can reduce effective sample size and introduce attrition bias not captured by the present simulation.
- Future simulation extensions should incorporate realistic compliance and dropout mechanisms.

The simulation focused on a single outcome measure (nightmare frequency) and did not consider potential differential effects on multiple endpoints.

- Multiple primary and secondary outcomes could affect the relative efficiency of different designs.
- Single-case and multi-outcome extensions of N-of-1 power analysis remain areas for further development.
- The present findings are therefore most directly applicable to designs with a single primary outcome.

The carryover decay is modeled as exponential in the DGP; alternative decay shapes are illustrated analytically but not yet simulated, and the robustness of power estimates to decay-model choice remains an open question.

- Linear and Weibull decay alternatives are pre-registered as sweep S3 and deferred.

- More complex carryover patterns involving neuroadaptation or learning effects may require modeling approaches not examined here.
- The present power estimates should be read as conditional on exponential decay.

8.6.2 Simulation Uncertainty

Following Morris et al. (2019), all performance measures are reported with Monte Carlo standard errors; at $n_{\text{sim}} = 2{,}000$, both power and Type I error MCSEs fall below the 1.5-percentage-point pre-registration target.

- The MCSE on a power estimate near 0.75 is approximately 1.0 percentage points.
- The MCSE on a Type I estimate near 0.05 is approximately 0.5 percentage points.
- MCSE bars in the power, bias, and coverage figures should be borne in mind when interpreting fine-grained differences between adjacent cells.

Type I error evaluation confirmed that both designs maintain appropriate error control at the nominal 5% level, supporting the validity of the power comparisons.

- Both designs' Type I confidence intervals include the nominal 5% level.
- This confirms that the observed power differences reflect genuine effect detection rather than inflated false-positive rates.
- Adequate Type I control is a prerequisite for interpreting the power advantage as substantively meaningful.

8.7 Comparison with Prior Simulation Studies

Five prior simulation studies provide direct quantitative benchmarks for the present results; comparisons with three additional studies are deferred pending full-text availability.

- Head-to-head comparisons with Blackston 2019, Hendrickson 2020, Chen/Chen/Xia 2014, Tang and Landes 2020, and Wang and Schork 2019 are completed from full-text sources.
- Comparisons with Shi and Ye, Pequignat et al., and Chen, Li, and Yang are deferred to the companion analysis.
- Quantitative benchmarks focus on scenarios where parameter ranges overlap with the present study.

8.7.1 Blackston et al. (2019): aggregated N-of-1 vs. parallel and crossover RCTs

Blackston et al. simulated 5,000 trials of aggregated N-of-1, parallel RCT, and 2-period crossover designs and reported the N-of-1 power and sample-size advantages along with Type I inflation under unmodeled carryover.

- Power at $n = 30$ was 92% for N-of-1, 32% for RCT; reaching 80% power required $n = 150$ for the RCT versus $n \leq 30$ for N-of-1.

- Under unmodeled carryover equal to 40% of the treatment effect, N-of-1 Type I inflated to 15%; at 60%, to 25%.
- Adjusting for carryover in the analysis model restored nominal Type I.

The sample-size advantage direction and magnitude from Blackston et al. are concordant with the present results; a principled analytical difference explains why the present simulation does not reproduce Blackston's Type I-error-under-unmodeled-carryover finding.

- Blackston's 2.5-fold reduction in required n is consistent with the present result of $N \leq 16$ for N-of-1 versus $N \approx 40$ for RCT.
- The present analysis always either excludes contaminated periods or includes a continuous `Dbc` regressor absorbing carryover, preventing Type I inflation.
- This is a principled analytical specification difference, not a disagreement on the underlying statistical behavior.

8.7.2 Chen, Chen, and Xia (2014): four analysis-method comparison

Chen, Chen, and Xia compared four analysis approaches across 3-cycle and 4-cycle aggregated N-of-1 trials, reporting Type I error control and power for compound-symmetry and AR(1) structures with and without carryover.

- Four analysis methods were compared: paired t-test, mixed differences model, full mixed model with carryover regressor, and DerSimonian-Laird meta-analysis.
- Three compound-symmetry covariance structures and two AR(1) structures were evaluated at $n \in \{1, 3, 5, 10, 20, 30\}$.
- Simulations used 3,000 replicates per cell with 0% or 20% carryover.

Under compound-symmetry structures, DerSimonian-Laird meta-analysis showed inflated Type I while other methods were near nominal; under carryover, only the mixed-effects model with explicit carryover term was unbiased.

- Under AR(1) structures, all models were slightly conservative (0.020 to 0.040).
- With 20% carryover, the M3 model with explicit carryover regressor was unbiased while M1, M2, and M4 power declined 1–10 percentage points.
- Chen and colleagues recommend the paired t-test as default and the mixed model with explicit carryover when carryover is present.

The present study extends Chen et al. by adding `corCAR1` residual serial-correlation structure, and the findings are broadly concordant on Type I control and carryover handling.

- Chen et al.'s finding that Type I is controlled under 20% carryover by the M3 family is concordant with the S12 null calibration result.
- Chen et al.'s finding that DerSimonian-Laird meta-analysis performs poorly at small n motivated replacement of M4 with the M3-family `lme/corCAR1` model.

- Period exclusion and the continuous `Dbc` regressor represent complementary rather than competing analytical strategies.

8.7.3 Tang and Landes (2020): serial t-tests for single-individual N-of-1

Tang and Landes developed serial t-tests correcting for AR(1) correlation at the single-individual level, demonstrating that uncorrected analyses inflate Type I and that power collapses under high positive serial correlation with short series.

- At $m = 12$, $\rho = 0.67$, the usual paired t-test achieves Type I of 0.22 while the serial t-test achieves 0.06.
- Effect sizes detectable with 80% power at $m = 4$ inflate from 1.65σ at $\rho = 0$ to 13.73σ at $\rho = 0.67$.
- These results demonstrate the critical importance of accounting for serial correlation in N-of-1 analyses.

At $m = 12$, $\rho = 0.67$, the serial t-test restores near-nominal Type I error, and Table 2 of Tang and Landes quantifies the dramatic inflation of detectable effect sizes under positive serial correlation.

- The usual paired t-test achieves Type I of roughly 0.22 while the serial t-test achieves roughly 0.06.
- The detectable μ_{A-B} at $m = 4$, $\rho = 0.67$ is 13.73σ, a 48-fold inflation relative to the uncorrelated case.
- At $m = 12$, the same three autocorrelation levels yield detectable effects of 0.52σ, 1.25σ, and 5.43σ.

The Tang and Landes results speak to single-subject analysis targets outside the aggregated N-of-1 scope; the aggregated hybrid absorbs within-patient correlation via `corCAR1`, producing concordant Type I behavior.

- The aggregated-level analogue finds hybrid power saturated across ρ because `corCAR1` absorbs within-patient correlation across 40 participants.
- Both papers predict that analyses ignoring positive residual correlation will inflate Type I.
- The extreme detectable-effect-size inflation at small m cautions against single-subject N-of-1 analysis with fewer than approximately ten observations.

8.7.4 Concordance synthesis and remaining gaps

Across the five completed head-to-head comparisons, design ranking and direction of effects are concordant with the present results on all key dimensions.

- Aggregated N-of-1 achieves higher power at matched total sample size (Blackston, Hendrickson).
- Period stacking and adequate cycle count mitigate serial-correlation power loss (Wang and Schork, Tang and Landes).

- Analyses ignoring carryover or serial correlation inflate Type I at the design level where those nuisance components operate (Blackston, Chen, Tang and Landes).

Three prior-literature comparisons remain incomplete due to full-text unavailability, and are queued in the companion analysis document for future revision.

- Shi and Ye on behavioral carryover in 2-period crossovers is pending.
- Pequignat et al. on idiographic power for rare-condition precision medicine is pending.
- Chen, Li, and Yang on aggregated N-of-1 versus parallel and crossover RCTs is pending.

8.8 Future Research Directions

Empirical validation through head-to-head comparisons of N-of-1 and RCT designs in actual clinical trials would strengthen the evidence base for design selection.

- Real-trial comparisons would expose compliance and feasibility challenges absent from simulation.
- The simulation findings provide testable predictions that could be evaluated against actual trial data.
- Such validation is a natural next step for establishing the external validity of the simulation results.

The development of adaptive N-of-1 trial designs that optimize crossover periods based on accumulating data, particularly Bayesian adaptive designs, could further improve efficiency.

- Bayesian approaches are well-suited for adaptive N-of-1 designs.
- The simulation framework developed here could readily be extended to evaluate Bayesian adaptive variants.
- Such extensions would further address the joint n-versus-k optimization problem.

Investigation of hybrid designs combining N-of-1 and traditional RCT elements could capture advantages of both approaches while mitigating respective limitations.

- Hybrid designs may be particularly appropriate for settings where both individual and population-level inference are required.
- Quantitative evaluation of specific hybrid configurations can draw on the simulation framework developed here.
- Appropriate design selection depends on the relative weight given to individual versus population-level inferential goals.

Patient and clinician acceptance of N-of-1 methodology in real-world settings will be important for implementation, as treatment burden, complexity, and interpretability may influence practical utility.

- Factors such as treatment burden and complexity cannot be captured by simulation studies.
- Interpretability of individual versus population-level results may differ between clinical stakeholders.
- Implementation research complementing the present simulation evidence is warranted.

8.9 Preliminary validation run

Before the production simulation was finalized, a preliminary validation run on the same 8-cell design slice verified computational feasibility and confirmed the direction of the main findings, while identifying two implementation issues.

- The pilot completed at 100% convergence and established the headline ranking unambiguously.
- Two implementation issues were identified: undercalibrated within-patient residual SD and failure to exclude carryover-contaminated periods.
- Both issues were corrected in the production run.

The within-patient residual SD in the initial run was too small relative to the target the mean-moderation architecture Gompertz DGP, yielding empirical SE approximately 0.107 versus the target 0.28.

- The production run addresses this by calibrating `SIGMA_WITHIN` to 2.3.
- This calibration yields empirical SE approximately 0.28 at the null cell.
- The undercalibrated SD produced artificially narrow effect-estimate distributions, inflating apparent power in the pilot.

The initial run did not exclude carryover-contaminated placebo periods, causing bias of $\hat{\beta}_D$ to grow with $|\beta_D|$ and coverage to collapse from 94.9% at the null to 37.5% at $\beta_D = -2$.

- The production run excludes contaminated periods using the `is_carryover` flag set by the data-generating process.
- Exclusion removes the bias and restores coverage to its nominal level.
- This finding confirms that period exclusion is necessary for unbiased estimation under the present DGP.

9 Conclusions

We have reported a simulation study comparing aggregated N-of-1 hybrid trials and parallel-group RCTs at matched participant count across a clinically calibrated range of effect sizes and carryover half-lives.

- Three principal conclusions are emphasized.
- The comparison is conducted under the ADEMP framework with Monte Carlo standard errors for all performance measures.

- Parameters are calibrated to the prazosin/PTSD reference application.

The N-of-1 hybrid achieved substantially higher power than the parallel-group RCT at every non-null effect size, with the largest advantage in the small-to-moderate effect regime where the parallel RCT reaches only 10–23% power.

- The N-of-1 hybrid achieved 31–83% power at $\beta_D \in [-1.0, -0.5]$ nightmares per week versus 10–23% for the RCT.
- The absolute power gap exceeded five combined MCSEs at every non-null cell.
- The ranking is not attributable to simulation noise.

Period exclusion for carryover management proved robust: power, bias, and empirical SE were essentially constant across all five carryover half-life cells, and bias was negligible with coverage tracking the nominal 95% level throughout.

- The fitted model operates only on non-contaminated periods after exclusion, making the true half-life irrelevant to the estimating equations.
- Robustness held across $t_{1/2} \in \{0.5, 1.0, 3.0, 5.0, 7.0\}$ days in the S1 sweep.
- Coverage was maintained at the nominal level for both designs at all effect-size cells.

The `lme/corCAR1` model with Satterthwaite denominator degrees of freedom provided adequate Type I error control at $N = 35$; the parallel-group RCT showed a marginally elevated rate warranting further investigation.

- N-of-1 Type I error was 0.051, within one MCSE of the nominal 0.05.
- RCT Type I error was 0.061, consistent with known small-sample effects in `lm` at $N = 35$.
- The marginally elevated RCT rate warrants further investigation before the final manuscript is submitted.

10 Acknowledgments

We thank the clinical researchers whose published work on prazosin for PTSD provided the empirical foundation for simulation parameters, and acknowledge the methodological contributions of the N-of-1 trials research community.

- Empirical parameter calibration relied on published prazosin/PTSD clinical trials.
- Analytical frameworks from the N-of-1 methodology literature provided the foundation for the simulation design.
- The N-of-1 research community's ongoing methodological contributions are acknowledged.

11 Data availability statement

Simulation data and code supporting the findings are openly available in the pmsimstats-ng project repository, with per-replicate Monte Carlo outputs and ADEMP-compliant summaries versioned under `analysis/data/`.

- Per-replicate Monte Carlo outputs are versioned under `analysis/data/`.
- Simulation drivers are located at `analysis/scripts/`.
- Exact paths for this manuscript are listed in the Reproducibility section.

12 Reproducibility

This manuscript was rendered with R 4.4.x inside the project's zzcollab Docker container, with package dependencies captured in `renv.lock` and the principal simulation packages listed.

- The zzcollab Docker container image hash is recorded in the `Dockerfile`.
- Principal packages include `nlme`, `parallel`, `data.table`, `furrr`, `purrr`, `rmarkdown`, and `bookdown`.
- Full reproducibility requires both the container image and the `renv.lock` package manifest.

Each simulation driver sets `set.seed(20260507)` at the top of the replicate loop and uses `parallel::mc.set.seed` inside `mclapply`, with per-replicate seeds derived from the master seed.

- The master seed is applied once at program entry.
- Per-replicate seeds derive from the master seed by `+ rep_idx`.
- This scheme permits exact reproduction of any replicate while maintaining independence across replicates.

The canonical production simulation output is at `analysis/data/derived_data/04-mainfx-productio` and the driver scripts, ADEMP plan, and manuscript source are fully documented.

- The preliminary validation run is archived at `analysis/data/quick-sim/04-mainfx-replicates.rds`.
- Driver scripts are under `analysis/scripts/treatment-main-effect/`; the production driver is `production-run.R`.
- The pre-registered ADEMP plan is at `analysis/scripts/treatment-main-effect/00-ademp-pre-regis`